

50.046 Cloud Computing and IoT

Project briefing

2026 Term 7

Assoc. Prof. Jihong Park



SINGAPORE UNIVERSITY OF
TECHNOLOGY AND DESIGN

Course project overview

- A group project that:
 - **Identify / formulate** a {societal | business} {problem | opportunity} that can be addressed by IoT
 - **Design** an end-to-end IoT-cloud solution (requirements, constraints, design space exploration, tech stack, design choice justification)
 - **Develop** a proof-of-concept prototype that demonstrates *selected* features

Form your group

- Each group consists of 4 students (recommended)
- Register your group's info here: [link](#)

Project: logistics

- Each group will need to appoint one representative to collect IoT gadgets
- Keep good of the gadgets and return by Week 14 (TBD)

Devices we will provide

- ESP32
 - MakePython ESP32 Dev Kit
 - ESP32 CAM Arduino Kits



- Raspberry Pi
 - with touch screen



Some links to the gadget

- <https://www.makerfabs.com/makepython-esp32-starter-kit.html>
- <https://www.wish.com/product/esp32-camera-arduino-kits-monitor-snapshot-face-detection-recognition-wifi-bluetooth-camera-module-with-128m-sd-card-usb-to-serial-cable-hc-sr501-sound-sensor-compatible-for-arduino-idetutorial-5e1c341229e786439a698b05>

Project management

- Start now
- Each member expects to spend ~ 30 hours on the project (spend them in 2-3 hours / chunk)
 - So each team has ~ 100 hours to tackle some interesting problem together!
 - Work as a team. Start from a small, specific setting. Keep iterating. Keep your working journal.

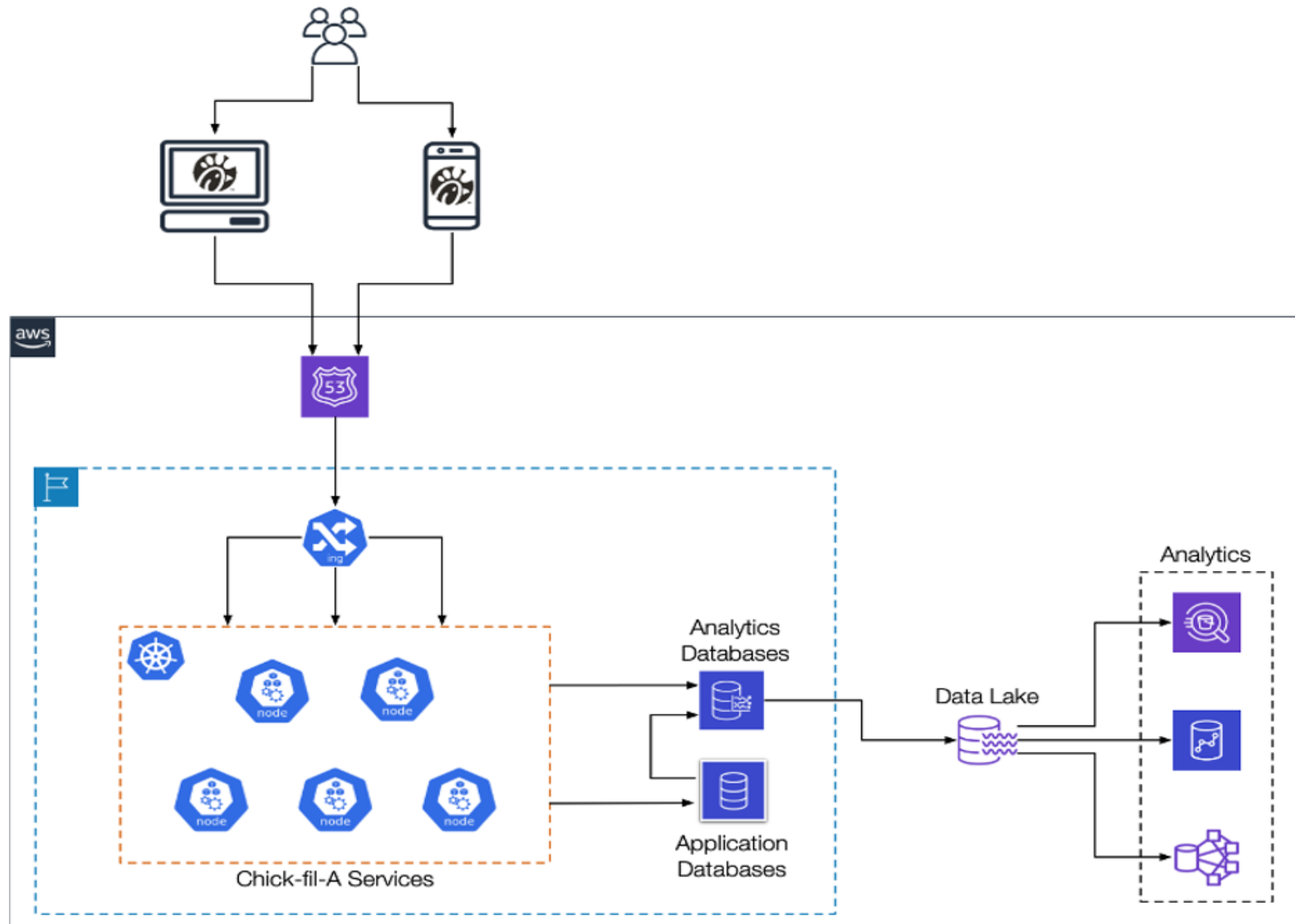
Some questions for you to consider

- What will be the benefit of my project? Can quantify in \$ or social impact?
- What features / use cases to support?
- What will be the cost? Quantify in \$ (CAPEX, OPEX, as a function of the system size, as a result of my design)
- Thing: What to sense? What (not) to sense? How powerful should be the MCU? What should / should not be included in my device?
- Connectivity: What / how / when to send? What protocol stack to use?
- Data: Any external data sources besides IoT data? How to design your data processing / storage pipeline?
- Additional key considerations? For example, real-time requirement / security / privacy / mobility / reliability / harsh environment? How my design address them?
- What kind of analytics? Where to put it? How to optimize it?
- How to deploy / bootstrap your system? How to maintain / upgrade your system? How to retire your system?
- How to interface with other services? What APIs you may want to provide?

Example I



Example II



Example III

